

Quiz 1

- Due No due date
- Points 10
- Questions 10
- Available Jan 16 at 7:10pm - Jan 18 at 11:59pm
- Time Limit None
- Allowed Attempts 3

Instructions

Intro and Universal Approximators

This quiz covers lectures 1 and 2. Several of the questions invoke concepts from the hidden slides in the slide deck, which were not covered in class. So please go over the slides before answering the questions.

You will have three attempts for the quiz. Questions will be shuffled and you will not be informed of the correct answers until after the deadline. While you may discuss the concepts underlying the questions with others, you must solve all questions on your own - see course policy.

[Take the Quiz Again](#)

Attempt History

	Attempt	Time	Score
KEPT	Attempt 2	14 minutes	8.5 out of 10
LATEST	Attempt 2	14 minutes	8.5 out of 10
	Attempt 1	326 minutes	8 out of 10

❗ Correct answers are hidden.

Score for this attempt: 8.5 out of 10

Submitted Jan 18 at 3:04pm

This attempt took 14 minutes.



Question 1

1 / 1 pts

Which of your quiz scores will be dropped?

Hint: watch Lecture 0

- ☐ Lowest 1 quiz scores
- ☒ Lowest 2 quiz scores
- ☐ Lowest 3 quiz scores
- ☐ No scores will be dropped



Question 2

1 / 1 pts

Which concept is most closely described by the following sentence? "The finding that neurons whose connections form cycles can model memory."

Hint: Lec 1, slide "other things MLPs can do"

- ☐ Sanger's Rule
- ☐ Turing's B-type machines
- ☐ Parallel Distributed Processing
- ☒ Lawrence Kubie's Loop Networks



Question 3

1 / 1 pts

Which of the following are **NOT** part of Aristotle's laws of association? (select all that apply)

Hint: See lec 1: Slides on "Associationism" 32-37

- ☒ Associations are formed from logical deduction.
- ☒ Mental willpower is used to associate unrelated events or things
- ☐ Thoughts about one event or thing triggers thoughts about related events or things.
- ☐ Thoughts about one event or thing triggers thoughts about events or things that have opposite qualities
- ☐ Events or things near the same place and time are associated.

In Lecture 1, Aristotle's laws of association are presented as basic mechanisms by which ideas become linked through experience. These laws focus on relationships such as proximity in time or space, similarity, contrast, and repeated co-occurrence. Importantly, they describe associations as arising naturally from how cognition works, rather than from deliberate reasoning or conscious effort. Options that appeal to mental willpower or logical deduction may sound plausible in everyday terms, but they fall outside the four associationist laws explicitly listed in the lecture.

Partial



Question 4

0.5 / 1 pts

Which of the following statements are true about MLPs? (select all that apply)

Slides: **Lec 2 slides: 153-154, 45-46**

☐

Even in theory, MLPs may need more than one hidden layer to compose complex decision boundaries of sufficient complexity.

☒

For linear/threshold units, the VC dimension of an MLP is bounded by the square of the number of weights in the network.

☒

Deeper networks generally require far fewer neurons than shallower networks to express the same function

☒

A network comprising exactly one hidden layer (plus the output layer) is a Universal Boolean Machine

For VC dimension, see lec 2, "Sufficiency of architecture". For deeper networks, lec 2, "The issue of depth". For universal boolean machines, lec 2, "How many layers for a Boolean MLP?"

For complex decision boundaries, theoretical MLPs only need one hidden layer to be universal approximators, although they may need near-infinite neurons. Depth reduces the total number of neurons needed.

Incorrect



Question 5

0 / 1 pts

If the first layer of a network operating on real-valued inputs does not have sufficient neurons to capture all the basic features of the function (map), which of the following activation functions gives subsequent layers the best chance to capture the features missed by the first layer.

Hint: Which activation function has the most gradation

Hint: Slides: Lec 2, "Sufficiency of architecture" slides 139 – 148

☐ Leaky Relu activation

☐ Threshold activation

☐ Relu activation

☒ Sigmoid activation

The threshold activation is saturated (to 0 or 1) immediately on either side of the boundary. The Sigmoid is more graded, but nonetheless saturates within a short interval. The RELU stays graded on one side of the boundary, but has a slope of 0 on the other, providing no information about the distance from the boundary on that side. The Leaky Relu has fixed non-zero slope on both sides of the boundary and continues to provide information about distance from the boundary on either side.



Question 6

1 / 1 pts

In general, as the depth of a NN increases, at what rate does the number of neurons/params required to represent a function change?

(Note: for the definition of network depth, see the lecture 2 recording)

Hint: Review Lec 2, slides on “The challenge of depth” (Slides 67-68)

- ☒ Decreases exponentially
- ☐ Increases quadratically
- ☐ Decreases linearly
- ☐ Increases linearly

The worst case problem of XORs is what we'll use for illustration. With one hidden layer you need $O(\exp(N))$ neurons. With 3 hidden layers, you need $O(\exp(N/2))$ neurons (the first 2 layers to compute XORs, which results in $N/2$ variables, and then one hidden layer to compute the XOR of $N/2$ variables. With 5 hidden layers the XOR net after K layers, the number of neurons required is $O(\exp(N/4))$ etc.

So, the reduction is exponential with increasing depth.

In general, for a given function, deeper networks will require exponentially fewer neurons (and hence parameters) than shallower ones to model the function accurately (exactly, or with arbitrary precision).



Question 7

1 / 1 pts

Although we haven't yet covered training of neural networks in the lectures, we can give you this advance bit of information: The number of required training inputs to train a network properly is monotonically related to the number of parameters.

In general, as depth increases, and because deeper networks require exponentially fewer parameters to represent the same function, how does the amount of training data required change?

Slide: lec 2, “The challenge of depth” slides 67-68

- ☒ Decreases exponentially
- ☐ Increases exponentially
- ☐ Decreases quadratically
- ☐ Increases quadratically

In general, for a given function, deeper networks will require exponentially fewer parameters than shallower ones to model the function accurately (exactly, or with arbitrary precision). Since the number of

required training inputs is monotonically related to the number of parameters required, the number of required training observations therefore decreases exponentially as depth increases.



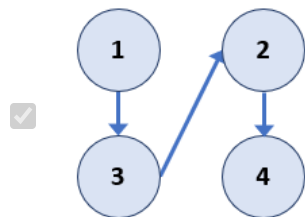
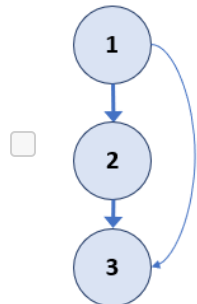
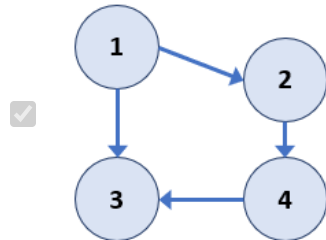
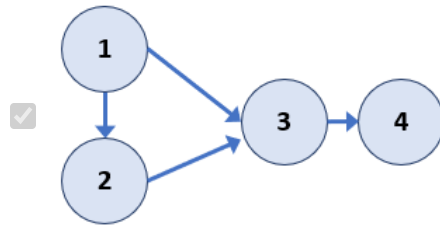
Question 8

1 / 1 pts

Which of the following NNs have a depth of 3? (Choose all that apply)

(Note: for the definition of network depth, see the lecture 2 recording)

Hint: lec 2, "Deep Structures", *Slides*: 17-18



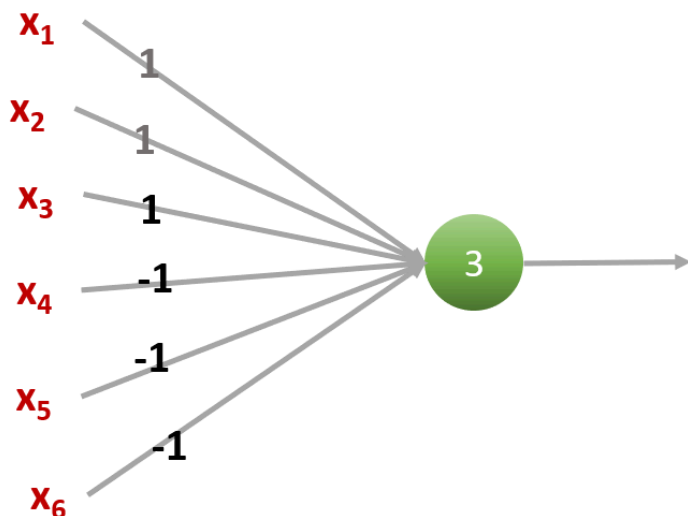
Depth is defined as longest path from source to sink.

- a) longest path: 1->2->3 (Depth 2)
- b) longest path: 1->2->4->3 (Depth 3)
- c) longest path 1->3->2->4 (Depth 3)
- d) longest path: 1->2->3->4 (Depth 3)



Question 9

1 / 1 pts



Under which condition(s) is the perceptron graph above guaranteed to fire? Note that $\sim$ is NOT. (select all that apply)

Slide: lec 2, "Perceptron as a Boolean gate" slides 26-30

- ☐ $\sim x_1 \& \sim x_2 \& \sim x_3 \& x_4 \& x_5 \& x_6$
- ☒ $x_1 \& x_2 \& x_3 \& \sim x_4 \& \sim x_5 \& \sim x_6$

$$x_1 \& x_2 \& x_3 \& \sim x_4 \& \sim x_5 \& \sim x_6 = 1(1) + 1(1) + 1(1) + 0(-1) + 0(-1) + 0(-1) = 3$$

$$x_1 \& \sim x_2 \& x_3 \& \sim x_4 \& x_5 \& \sim x_6 = 1(1) + 0(1) + 1(1) + 0(-1) + 1(-1) + 0(-1) = 1$$

$$\sim x_1 \& \sim x_2 \& \sim x_3 \& x_4 \& x_5 \& x_6 = 0(1) + 0(1) + 0(1) + 1(-1) + 1(-1) + 1(-1) = -3$$

- ☐ $x_1 \& \sim x_2 \& x_3 \& \sim x_4 \& x_5 \& \sim x_6$
- ☐ Never fires

For this perceptron to fire, you need the total to be ≥ 3 . Clearly you'll need all possible positive contributions, with all negative inputs turned off

$$x_1 \& x_2 \& x_3 \& \sim x_4 \& \sim x_5 \& \sim x_6 = 1(1) + 1(1) + 1(1) + 0(-1) + 0(-1) + 0(-1) = 3$$

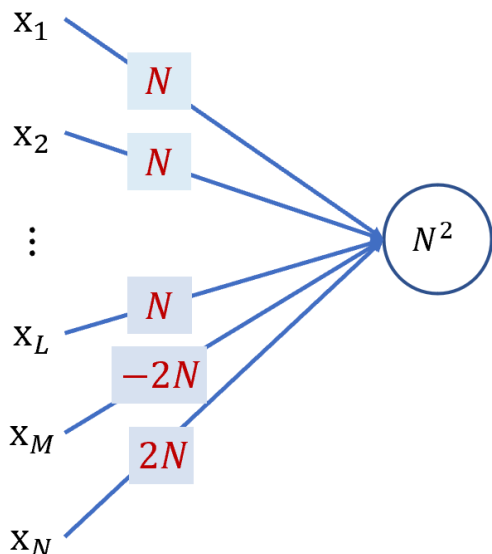
$$x_1 \& \sim x_2 \& x_3 \& \sim x_4 \& x_5 \& \sim x_6 = 1(1) + 0(1) + 1(1) + 0(-1) + 1(-1) + 0(-1) = 1$$

$$\sim x_1 \& \sim x_2 \& \sim x_3 \& x_4 \& x_5 \& x_6 = 0(1) + 0(1) + 0(1) + 1(-1) + 1(-1) + 1(-1) = -3$$

⋮

Question 10

1 / 1 pts



What boolean function does the above perceptron represent?

(Note: In this pic, $M = N - 1$ and $L = N - 2$. Notice that there are only dots between x_2 and x_L)

Slide: lec 2, "Perceptron as a Boolean gate" Slides 26-30

- ☐ $\left(\bigvee_{i=1}^L X_i \right) \wedge (X_M) \vee (\overline{X_N})$
- ☐ $\left(\bigwedge_{i=1}^L X_i \right) \vee (X_M) \wedge (\overline{X_N})$
- ☐ $\left(\bigvee_{i=1}^L X_i \right) \vee (\overline{X_M}) \vee (X_N)$
- ☒ $\left(\bigwedge_{i=1}^L X_i \right) \wedge (\overline{X_M}) \wedge (X_N)$

All inputs except X_M have positive weights, while X_M has a negative weight. There are N weights in total, also notice that $L = N - 2$. To cross the threshold, all weights other than X_M need to fire. This can be summarized as the boolean equation above, where the upside-down $\vee$ is an "and" symbol, and the bar above " X_M " indicates that it's negated.

Quiz Score: 8.5 out of 10